

WASSERKUPPE, Germany

WMO: 105440

Lat: 50.4972N

Lon: 9.9428E

Elev: 926

StdP: 90.69

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.7	-12.0	-18.5	0.8	-10.8	-16.1	1.0	-9.6	18.5	-0.1	16.6	-2.5	6.9	130	0.809

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	24.1	16.6	22.3	15.8	20.7	14.9	17.6	22.5	16.6	21.2	15.6	19.7	5.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	12.5	20.0	14.7	11.7	18.5	13.8	11.0	17.3	53.2	22.2	49.9	21.3	46.8	19.5	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.3	13.5	12.0	DB	-15.2	28.2	4.7	1.7	-18.6	29.4	-21.3	30.4	-23.9	31.3	-27.3	32.5	(4)
(5)			WB	-15.6	19.9	4.5	0.9	-18.8	20.6	-21.5	21.1	-24.0	21.6	-27.2	22.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-2.6	-3.1	0.3	3.6	9.0	11.7	14.2	14.2	10.5	6.5	1.4	-1.0
	DBStd	7.42	4.60	5.04	4.05	4.16	4.28	4.18	3.87	3.78	3.64	3.88	4.19	3.67
	HDD10.0	2127	391	368	302	194	73	29	7	7	37	121	259	340
	HDD18.3	4722	649	601	559	441	289	203	136	136	235	366	508	598
	CDD10.0	468	0	0	1	3	42	79	138	138	52	14	1	0
	CDD18.3	20	0	0	0	0	0	3	9	8	0	0	0	0
	CDH23.3	87	0	0	0	0	0	8	33	46	0	0	0	0
	CDH26.7	11	0	0	0	0	0	1	2	8	0	0	0	0
Wind	WSAvg	6.2	7.3	7.0	6.7	6.0	5.9	5.2	5.2	5.1	5.7	6.9	6.7	6.8
Precipitation	PrecAvg	880	81	68	64	51	73	71	89	74	69	72	79	90
	PrecMax	1089	185	193	130	97	163	170	158	195	159	232	125	246
	PrecMin	613	3	21	15	3	10	25	34	19	19	13	2	21
	PrecStd	132	40	39	34	22	41	32	31	35	36	43	32	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	10.2	14.4	16.4	21.5	24.4	26.5	27.7	22.2	19.1	13.1	8.9
		MCWB	3.6	5.7	7.6	9.8	13.6	16.8	18.6	18.3	15.1	12.8	7.4	6.3
	2%	DB	5.4	7.2	10.1	13.9	19.4	22.2	24.1	24.2	20.0	15.9	10.1	6.8
		MCWB	2.7	4.9	5.7	8.4	12.3	15.6	16.6	16.7	14.3	11.7	6.3	4.9
	5%	DB	3.8	5.1	7.6	12.0	17.7	20.1	22.2	22.2	18.1	13.6	8.5	5.5
		MCWB	2.4	2.9	5.0	7.2	11.6	14.5	15.9	16.0	13.7	10.4	6.3	3.8
	10%	DB	2.7	3.0	5.4	9.9	15.7	18.1	20.5	20.1	15.9	11.7	7.1	3.9
		MCWB	1.9	1.7	3.9	6.4	10.6	13.3	15.1	14.9	12.6	9.7	5.6	2.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	7.0	8.4	10.5	14.8	18.2	19.3	19.3	16.9	14.1	9.1	7.2
		MCDB	6.2	8.4	11.6	15.0	18.8	23.2	24.8	26.0	20.6	17.6	10.3	7.7
	2%	WB	3.8	5.2	6.7	9.1	13.2	16.3	17.7	17.5	15.1	12.3	7.9	5.9
		MCDB	4.2	6.8	9.4	12.8	17.7	20.7	22.3	22.8	18.4	14.5	8.9	6.5
	5%	WB	2.9	3.4	5.3	7.9	12.1	15.2	16.6	16.4	14.1	11.1	6.9	4.4
		MCDB	3.3	4.6	6.9	11.0	16.7	18.9	21.1	21.2	16.5	12.9	8.1	5.0
	10%	WB	1.9	1.9	4.0	6.7	11.1	14.0	15.5	15.3	13.1	10.0	5.7	3.2
		MCDB	2.5	2.7	5.4	9.7	15.1	17.4	19.7	19.2	15.4	11.4	6.8	3.5
Mean Daily Temperature Range	5% DB	MDBR	4.1	4.6	4.8	6.4	7.2	6.7	7.3	7.1	6.2	5.3	4.2	4.0
		MCDBR	6.2	7.4	7.8	9.2	10.0	9.3	9.5	10.0	8.8	8.1	6.4	5.8
		MCWBR	5.0	5.0	5.0	5.4	5.7	5.5	5.5	5.7	5.0	5.0	4.8	4.3
	5% WB	MCDBR	5.4	6.9	6.9	8.3	9.1	8.7	9.2	9.4	7.7	6.5	6.0	4.7
		MCWBR	5.0	5.1	4.9	5.3	5.6	5.5	5.8	5.7	5.1	4.6	5.0	4.2
Clear-Sky Solar Irradiance	taub	0.275	0.295	0.325	0.356	0.361	0.361	0.370	0.365	0.337	0.318	0.299	0.274	
	taud	2.461	2.427	2.404	2.351	2.353	2.363	2.330	2.370	2.437	2.494	2.497	2.460	
	Ebn at Noon	781	850	881	886	895	895	879	867	857	812	742	725	
	Edh at Noon	58	77	95	112	117	117	119	109	91	72	56	50	
All-Sky Solar Radiation	RadAvg	0.77	1.48	2.58	4.10	4.84	5.34	5.06	4.36	3.13	1.83	0.84	0.57	
	RadStd	0.12	0.22	0.36	0.52	0.55	0.48	0.61	0.42	0.44	0.21	0.12	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page